

Four problems, and the scaffolding falls away

A repressor gate, input x and output y , both as concentrations:

$$y(x) = y_{\min} + \frac{y_{\max} - y_{\min}}{1 + (x/K)^n}$$

Today's three tools, and nothing else. The gain is the log slope, and for this gate

$$G = \frac{d \ln y}{d \ln x}, \quad |G| = g \frac{y - y_{\min}}{y}, \quad g = \frac{n u^n}{1 + u^n}, \quad u = \frac{x}{K}.$$

The two solutions of $|G| = 1$ are the input thresholds $x_{IL} < x_{IH}$, and they name the worst-case output levels $y_{OH} = y(x_{IL})$ and $y_{OL} = y(x_{IH})$. When gate A drives gate B,

$$\text{NM}_L = \log_{10} \frac{x_{IL}(B)}{y_{OL}(A)}, \quad \text{NM}_H = \log_{10} \frac{y_{OH}(A)}{x_{IH}(B)},$$

and both must be positive. At each transition this sheet asks *why does that step follow?* Answer in writing before you go on.

1 — Fully worked: one gate, $n = 2$

Take $K = 1$, $y_{\min} = 0.1$, $y_{\max} = 10$ (so a hundredfold swing). Setting $|G| = 1$ and solving numerically:

$$x_{IL} = 1.02, \quad x_{IH} = 9.80,$$

$$y_{OH} = y(1.02) = 4.95, \quad y_{OL} = y(9.80) = 0.20.$$

How those two roots were found, since $|G| = 1$ does not factor: evaluate $|G|$ on a grid of x , find where it peaks (here $|G| = 1.64$ at $x = 3.2$), and then bisect on each side of the peak for the crossing. That is what `posb.digital.Repressor.thresholds` does, and it raises an error rather than returning numbers when the peak never reaches 1.

Those four numbers are the gate's datasheet. *Why does that step follow?* — the peak $|G|$ is above 1, so $|G| = 1$ has two roots. A gate whose peak gain never reached 1 would have none, and no digital reading at all.

2 — One step given: the same gate at $n = 3$

Same K , same $y_{\min}$ and $y_{\max}$, cooperativity 3 instead of 2. The peak $|G|$ is now 2.45, at $x = 2.15$, and $x_{IL} = 0.80$ is given.

$$x_{IH} = \underline{\hspace{2cm}} \quad (\text{it is between 5 and 6})$$

$$y_{OH} = y(x_{IL}) = \underline{\hspace{2cm}}, \quad y_{OL} = y(x_{IH}) = \underline{\hspace{2cm}}$$

Why does that step follow? — compared with item 1, did the band between the thresholds get wider or narrower, and is that good or bad for a designer?

3 — Nothing given: does this pair connect?

Gate A is item 1's gate, $n = 2$. Gate B is item 2's gate, $n = 3$ — same K , same leak, sharper. **A drives B**.
Use your own numbers from items 1 and 2.

$$NM_L = \log_{10} \frac{x_{IL}(B)}{y_{OL}(A)} = \log_{10} \frac{\underline{\hspace{1cm}}}{\underline{\hspace{1cm}}} = \underline{\hspace{1cm}}$$

$$NM_H = \log_{10} \frac{y_{OH}(A)}{x_{IH}(B)} = \log_{10} \frac{\underline{\hspace{1cm}}}{\underline{\hspace{1cm}}} = \underline{\hspace{1cm}}$$

One of the two fails, and only just. You may change A's ribosome binding site, which multiplies $y_{\min}$ and $y_{\max}$ by the same factor f . What range of f makes both margins positive?

f from to

Why does that step follow? — say in one sentence why changing A's RBS does not move A's own thresholds.

4 — Bare problem: a sensor that must read four decades

A collaborator wants a reporter whose output tells them the inducer concentration anywhere from 1 nM to 10 μ M — four decades — not merely whether it is above or below a threshold.

Explain why no single Hill-type promoter can do this, using a number. Then name a change to the circuit that would widen the usable range, and say which saturation your change removes.

If you finish early. Your gate from item 2 has $n = 3$. Over what fold-range of input is it a usable analog sensor, and how does that compare with item 1's gate?
